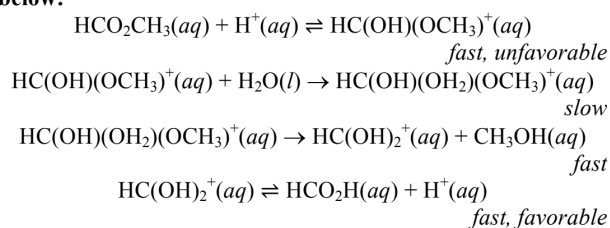
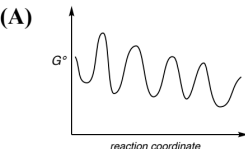
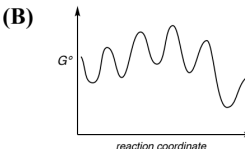
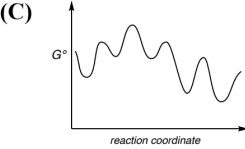
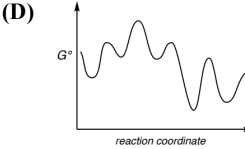


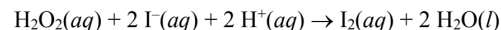
Kinetics

- The irreversible reaction of a compound A takes place with $\text{Rate} = k[\text{A}]^n$. The half-life of the reaction is independent of $[\text{A}]$. What is the value of n ?
 (A) 0
 (B) 1
 (C) 2
 (D) It cannot be determined from the information given.
- An acid-catalyzed reaction that is first-order in H^+ has a rate of $5.0 \times 10^{-6} \text{ M s}^{-1}$ when carried out in a buffer with $\text{pH} = 5.0$. What is the rate when the reaction is carried out with all reactant concentrations the same, but with the pH buffered at 5.5?
 (A) $1.6 \times 10^{-6} \text{ M s}^{-1}$ (B) $4.5 \times 10^{-6} \text{ M s}^{-1}$
 (C) $5.5 \times 10^{-6} \text{ M s}^{-1}$ (D) $1.6 \times 10^{-5} \text{ M s}^{-1}$
- The decomposition of hydrogen peroxide to form water and oxygen in the presence of an iodide catalyst is studied, and the rate law is found to be $\text{Rate} = k[\text{H}_2\text{O}_2][\text{I}^-]$ with $k = 1.05 \text{ M}^{-1} \text{ min}^{-1}$. A reaction is carried out with the initial concentrations of both iodide and hydrogen peroxide equal to 0.300 M. What is the concentration of hydrogen peroxide after 5.00 min?
 (A) 0.00157 M (B) 0.0621 M
 (C) 0.117 M (D) 0.145 M
- The temperature dependence of the rate constant for the gas-phase reaction of acrolein with nitrate radical is measured, and the rate constant is found to be $k = (1.72 \times 10^{-13})e^{-1190/T}$, with k in units of $\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ and T in units of K. What is the activation energy of the reaction?
 (A) 1.19 kJ mol^{-1} (B) 9.89 kJ mol^{-1}
 (C) 36.4 kJ mol^{-1} (D) 143 kJ mol^{-1}

Questions 5-6 concern the mechanism of acid-catalyzed hydrolysis of methyl formate in aqueous solution shown below:



- What rate law is predicted by this mechanism?
 (A) $\text{Rate} = k[\text{HCO}_2\text{CH}_3]$
 (B) $\text{Rate} = k[\text{HCO}_2\text{CH}_3][\text{H}^+]$
 (C) $\text{Rate} = k[\text{HCO}_2\text{CH}_3]/[\text{H}^+]$
 (D) $\text{Rate} = k[\text{H}^+]$
- Which reaction coordinate diagram best corresponds to the proposed mechanism?
 (A)  (B) 
 (C)  (D) 
- The rate law for the oxidation of iodide to iodine by hydrogen peroxide in weakly acidic solution is studied by allowing the reaction to proceed in the presence of a fixed amount of sodium thiosulfate (which rapidly reduces iodine to iodide) and measuring the time t to the first appearance of a blue color due to the starch-iodine complex. The following data are obtained on solutions made with the indicated volumes of reactant solutions and enough water to give a total volume of 10.0 mL. What are the reaction orders in H_2O_2 , I^- , and H^+ ?

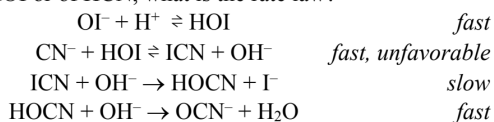


Run	CH_3COOH , mL	NaOH , mL	KI , mL	H_2O_2 , mL	t , s
1	2.0	1.0	2.0	2.0	68.2
2	4.0	1.0	2.0	2.0	68.9
3	2.0	1.0	4.0	2.0	33.2
4	2.0	1.0	2.0	4.0	32.9

	Order in H_2O_2	Order in I^-	Order in H^+
(A)	0	1	1
(B)	1	1	0
(C)	1	1	1
(D)	1	2	2

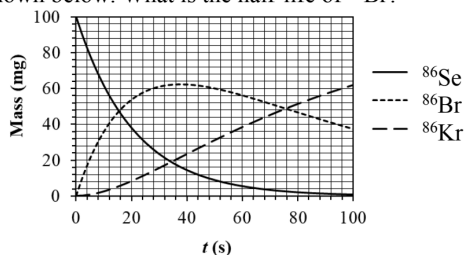
Kinetics

8. The oxidation of cyanide ion to cyanate ion by hypoiodite is proposed to take place by the following mechanism. If the pH is greater than either the pK_a of HOI or of HCN, what is the rate law?



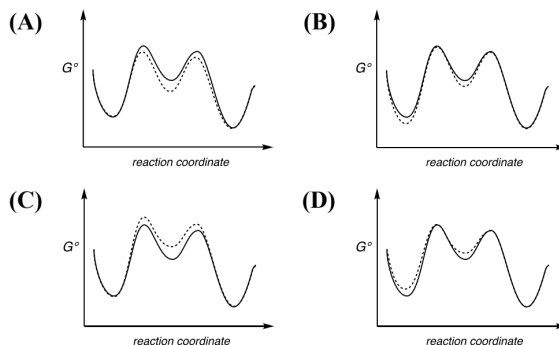
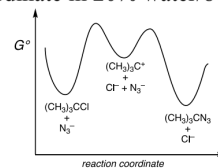
- (A) $\text{Rate} = k[\text{CN}^-][\text{OI}^-]$
 (B) $\text{Rate} = k[\text{CN}^-][\text{OI}^-][\text{H}^+]$
 (C) $\text{Rate} = k[\text{CN}^-][\text{OI}^-][\text{OH}^-]$
 (D) $\text{Rate} = k[\text{CN}^-][\text{OI}^-][\text{OH}^-]^2$

9. ^{86}Se undergoes β^- decay to ^{86}Br with a half-life of 14.3 s. ^{86}Br then undergoes β^- decay to ^{86}Kr , a stable product. 0.100 g ^{86}Se was placed in a sealed vessel, and the masses of all three species were recorded over time as shown below. What is the half-life of ^{86}Br ?

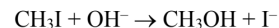


- (A) 55 s
 (B) 70. s
 (C) 85 s
 (D) It cannot be determined from the information given.
10. A reactant **R** undergoes independent, irreversible first-order reactions to form two products, **X** and **Y**. The reaction to form **X** has an activation energy that is 20.0 kJ mol^{-1} higher than the activation energy of the reaction to form **Y**. At 50.0 °C, the yield of **X** is 50.0%. What is the yield of **X** at 70.0 °C?
- (A) 32.5% (B) 39.4%
 (C) 60.6% (D) 77.0%

11. The reaction of *tert*-butyl chloride, $(\text{CH}_3)_3\text{CCl}$, with azide ion placed via the carbocation intermediate $(\text{CH}_3)_3\text{C}^+$ as shown on the reaction coordinate diagram below. The reaction proceeds faster in solvents of high polarity (such as water) than in less polar solvents (such as ethanol). The solid line represents the reaction coordinate in 80% water/20% ethanol. Which dashed line best represents the reaction coordinate in 20% water/80% ethanol?



12. The reaction of hydroxide ion with methyl iodide takes place in a single elementary step. The reaction is faster when the carbon-12 isotope is present than when carbon-14 is present ($k_{^{12}\text{C}}/k_{^{14}\text{C}} = 1.088$). What is the principal reason for the difference in rates?



- (A) The carbon nucleus is less stable in $^{14}\text{CH}_3\text{I}$.
 (B) The vibrational energies are lower in $^{14}\text{CH}_3\text{I}$.
 (C) The molecule is less sterically hindered in $^{14}\text{CH}_3\text{I}$.
 (D) The carbon-iodine bond dissociation enthalpy is smaller in $^{14}\text{CH}_3\text{I}$.

13. For a reaction $\text{X} + \text{Y} \rightarrow \text{Z}$, data for three experiments are given.

$[\text{X}]_0, \text{M}$	$[\text{Y}]_0, \text{M}$	Initial rate, $\text{M}^{-1} \text{min}^{-1}$
0.10	0.10	2.0×10^{-4}
0.30	0.10	6.0×10^{-4}
0.30	0.30	5.4×10^{-3}

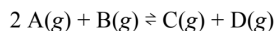
Which mechanism is consistent with the above data? (Reactions shown as reversible are assumed to be rapid and unfavorable.)

- (A) $\text{X} + \text{Y} \rightarrow \text{Z}$ (B) $2 \text{X} \rightleftharpoons \text{X}_2$
 $\text{X}_2 + \text{Y} \rightarrow \text{Z} + \text{X}$
 (C) $2 \text{Y} \rightleftharpoons \text{Y}_2$ (D) $\text{X} + \text{Y} \rightleftharpoons \text{XY}$
 $\text{Y}_2 + \text{X} \rightarrow \text{Z} + \text{Y}$ $\text{XY} \rightarrow \text{Z}$

14. ^{208}Po undergoes α decay to form stable ^{204}Pb with a half-life of 2.90 years. A sample of ^{208}Po was measured with a Geiger counter 5.20 years ago as having an activity of 1320 disintegrations per second. What is its activity today?

(A) 137 s^{-1} (B) 381 s^{-1}
(C) 736 s^{-1} (D) 1320 s^{-1}

15. A reversible reaction occurs as follows:



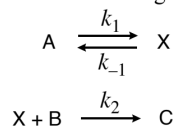
Under a certain set of conditions, the rate law for the forward reaction is determined to be $\text{Rate} = k_f[\text{A}][\text{B}]$. Under these conditions, what is the rate law for the reverse reaction?

- (A) $\text{Rate} = k_r[\text{C}][\text{D}]$
(B) $\text{Rate} = \frac{k_r[\text{C}][\text{D}]}{[\text{A}]}$
(C) $\text{Rate} = \frac{k_r[\text{C}][\text{D}]}{[\text{A}]^2[\text{B}]}$
(D) It cannot be determined from the information given.

16. A compound A reacts by two independent irreversible pathways to give two products, B and C. At 300 K, 35% of the product is B, while at 320 K, 70% of the product is B. What may be concluded about the activation energy of the pathway to form B compared to the activation energy of the pathway to form C?

- (A) The E_a for the pathway to form B is higher by 28 kJ mol^{-1} .
(B) The E_a for the pathway to form B is higher by 59 kJ mol^{-1} .
(C) The E_a for the pathway to form B is lower by 28 kJ mol^{-1} .
(D) The E_a for the pathway to form B is lower by 59 kJ mol^{-1} .

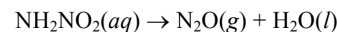
17. The irreversible reaction $\text{A} + \text{B} \rightarrow \text{C}$ takes place by the following two-step mechanism involving an intermediate X whose concentration is low throughout the reaction:



Which statement cannot be correct regardless of the values of k_1 , k_{-1} , and k_2 ?

- (A) The reaction is 0th order in A.
(B) The reaction is 1st order in A.
(C) The reaction is 0th order in B.
(D) The reaction is 1st order in B.

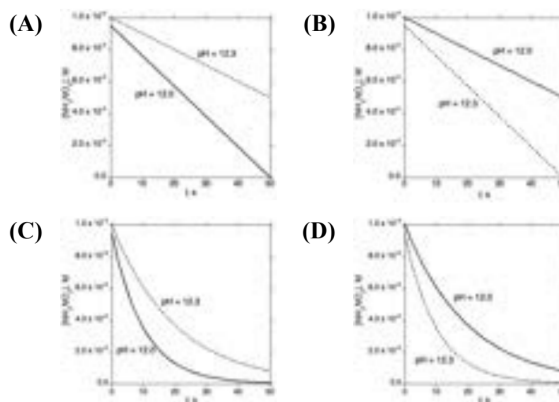
18. Nitramide decomposes to nitrous oxide in aqueous solution:



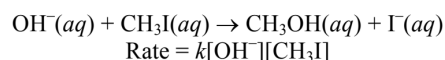
In solutions with $\text{pH} \geq 12$, the rate law for this reaction is:

$$\text{Rate} = k \frac{[\text{NH}_2\text{NO}_2]}{[\text{H}_3\text{O}^+]}$$

The reaction is monitored with $\text{pH} = 12.0$ (solid line) and with $\text{pH} = 12.3$ (dashed line). Which graph best represents the time-dependence of the nitramide concentrations in the two experiments?



19. Hydroxide ion reacts with methyl iodide to give methanol in an irreversible second-order reaction with a rate constant $k = 6.5 \times 10^{-5}\text{ L mol}^{-1}\text{ s}^{-1}$.



How long will it take the concentration of CH_3I to decrease from 0.010 M to 0.0020 M in a 1.00 M solution of sodium hydroxide?

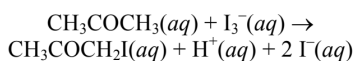
- (A) 120 s (B) $2.5 \times 10^4\text{ s}$
(C) $4.1 \times 10^6\text{ s}$ (D) $6.2 \times 10^6\text{ s}$

20. The temperature-dependence of the rate constant of a chemical reaction can be described by the Arrhenius equation, $k = Ae^{-E_a/RT}$. Which elementary chemical reaction has the largest Arrhenius pre-factor A?

- (A) $(\text{CH}_3)_3\text{CBr}(\text{aq}) \rightarrow (\text{CH}_3)_3\text{C}^+(\text{aq}) + \text{Br}^-(\text{aq})$
(B) $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$
(C) $\text{CH}_3\text{NC}(\text{g}) \rightarrow \text{CH}_3\text{CN}(\text{g})$
(D) $(\text{CH}_3\text{O})_2\text{SO}_2(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CH}_3\text{OH}(\text{aq}) + \text{CH}_3\text{OSO}_3^-(\text{aq})$

Kinetics

21. The iodination of acetone is irreversible and has a rate law of $\text{Rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+]$.

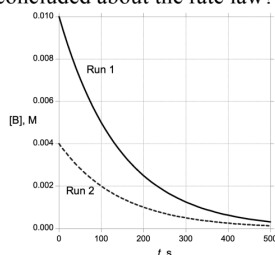


When the reaction is carried out with an initial concentration of I_3^- much smaller than those of either acetone or H^+ , it is observed that the yellow color due to I_3^- fades over time, finally disappearing abruptly at a time t . How does t depend on the initial concentrations?

Dependence on: $[\text{CH}_3\text{COCH}_3]$ $[\text{H}^+]$ $[\text{I}_3^-]$

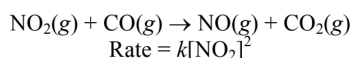
- (A) Direct Direct None
(B) Inverse Inverse None
(C) Direct Direct Inverse
(D) Inverse Inverse Direct

22. The reaction between A and B is carried out twice, with $[\text{A}]_0 = 0.50 \text{ M}$ in each run but with two different initial concentrations of B, and the results are shown below. What can be concluded about the rate law?



- (A) $\text{Rate} = (0.0069 \text{ s}^{-1})[\text{B}]$
(B) $\text{Rate} = (0.0069 \text{ M}^{-1} \text{ s}^{-1})[\text{A}][\text{B}]$
(C) $\text{Rate} = (0.014 \text{ M}^{-1} \text{ s}^{-1})[\text{A}][\text{B}]$
(D) The reaction is first order in B, but the order in A cannot be determined from the available data.

23. Nitrogen dioxide reacts with carbon monoxide with the rate law shown

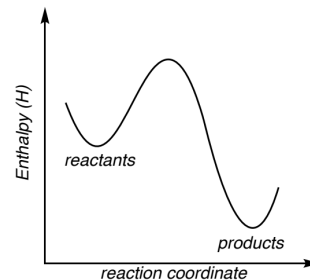


Which mechanisms are consistent with this rate law?

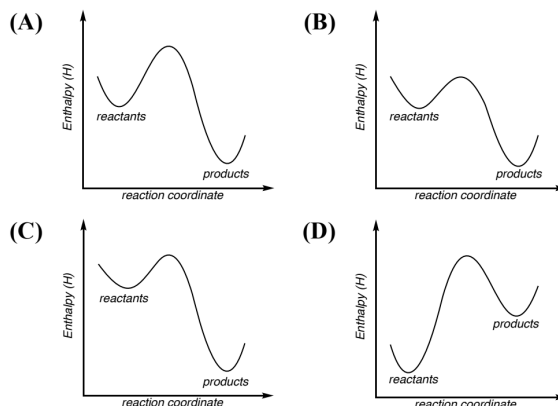
- I. $2 \text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$ *fast, unfavorable*
 $\text{N}_2\text{O}_4 + \text{CO} \rightarrow \text{N}_2\text{O}_3 + \text{CO}_2$ *slow*
 $\text{N}_2\text{O}_3 \rightarrow \text{NO} + \text{NO}_2$ *fast*
 II. $2 \text{NO}_2 \rightarrow \text{NO} + \text{NO}_3$ *slow*
 $\text{NO}_3 + \text{CO} \rightarrow \text{NO}_2 + \text{CO}_2$ *fast*

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

24. The reaction coordinate diagram of a reaction that takes place at 300 K is shown below.



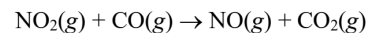
Which best represents the reaction coordinate diagram for the same reaction at 340 K?



25. The isotope actinium-226 undergoes both beta decay, with a half-life of 35.4 d, and electron capture, with a half-life of 173 d. What is the overall half-life for the radioactive decay of ^{226}Ac ?

- (A) 29.4 d (B) 78.3 d (C) 104 d (D) 208 d

26. The rate of the gas-phase reaction of nitrogen dioxide with carbon monoxide at 177 °C was measured under three different sets of concentrations as shown. What is the rate law for this reaction under these conditions?



	Initial $[\text{NO}_2]$, mol L^{-1}	Initial $[\text{CO}]$, mol L^{-1}	Initial rate, $\text{mol L}^{-1} \text{min}^{-1}$
Ex 1	4.0×10^{-4}	1.7×10^{-2}	1.70×10^{-7}
Ex 2	4.0×10^{-4}	3.4×10^{-2}	1.70×10^{-7}
Ex 3	1.2×10^{-3}	3.4×10^{-2}	1.53×10^{-6}

- (A) $\text{Rate} = k[\text{NO}_2]$ (B) $\text{Rate} = k[\text{NO}_2]^2$
(C) $\text{Rate} = k[\text{CO}]$ (D) $\text{Rate} = k[\text{NO}_2][\text{CO}]$

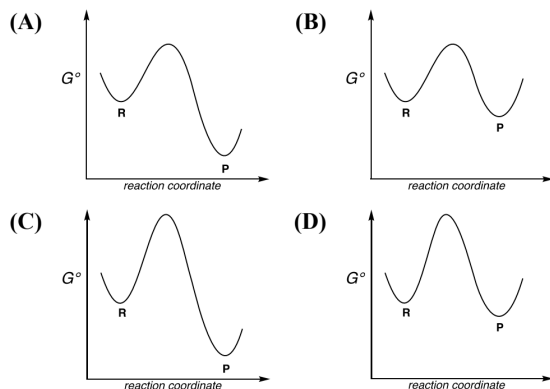
27. Which statement best describes the properties of the activation energy of a reaction?
- (A) It increases in the presence of a catalyst.
 - (B) It decreases with increasing temperature.
 - (C) It must be larger than the energy change for forming any intermediates in the reaction.
 - (D) It must be positive for endothermic reactions and negative for exothermic reactions.

28. The kinetics of the reaction of crystal violet with excess sodium hydroxide to form colorless products is monitored using a spectrophotometer. A plot of the natural logarithm of the absorbance of green light by the system as a function of time is linear with a negative slope. Which conclusions may be drawn from the data?

- I. The reaction is first-order in crystal violet.
- II. The reaction is first-order in hydroxide ion.

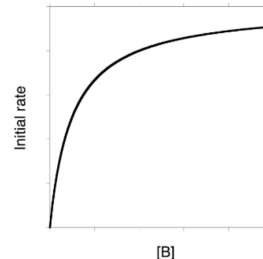
- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

29. A reaction initially contains pure reactant **R**, which forms product **P** reversibly. Which reaction coordinate diagram corresponds to the reaction in which **P** achieves a concentration within 5% of its equilibrium value the fastest?



Continued on next page

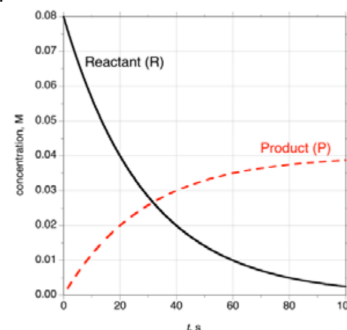
30. The reaction between **A** and **B** involves an intermediate **I** whose concentration remains very low throughout the reaction. The initial rates of a reaction as a function of **[B]** (at constant **[A]**) are shown below. Which statement about the mechanism is most consistent with the data?



- (A) **A** initially reacts reversibly to form the intermediate **I**, which then reacts irreversibly with **B** in a process that is first-order in **B**.
- (B) **A** initially reacts reversibly to form the intermediate **I**, which then reacts irreversibly with **B** in a process that is second-order in **B**.
- (C) **B** initially reacts reversibly in a first-order process to form the intermediate **I**, which then reacts irreversibly with **A**.
- (D) **B** initially reacts reversibly in a second-order process to form the intermediate **I**, which then reacts irreversibly with **A**.

31. The time course of a reaction in which a reactant **R** is converted to a product **P** is shown below. Which statements about the reaction are correct?

- I. The reaction is first order in **R**.
- II. Two moles of **P** are formed for every mole of **R** that reacts.



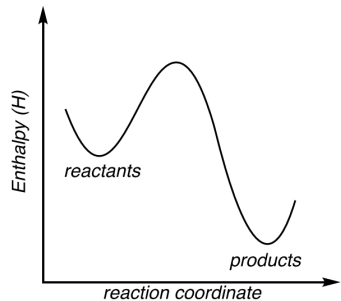
- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

Kinetics

32. In a reaction $A + 2 B \rightarrow C + D$, initial rates were measured as shown in the table. What is the initial rate of the reaction with $[A]_0 = 0.200 \text{ M}$ and $[B]_0 = 0.400 \text{ M}$?

$[A]_0, \text{M}$	$[B]_0, \text{M}$	Initial rate, M s^{-1}
0.120	0.300	1.80×10^{-5}
0.180	0.450	4.05×10^{-5}
0.080	0.200	8.00×10^{-6}
0.200	0.400	???

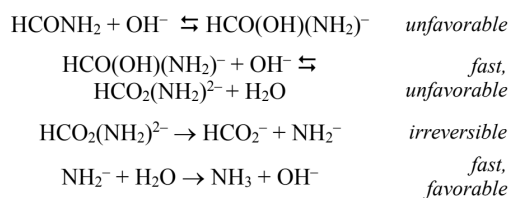
- (A) $3.20 \times 10^{-5} \text{ M s}^{-1}$
 (B) $4.00 \times 10^{-5} \text{ M s}^{-1}$
 (C) $5.00 \times 10^{-5} \text{ M s}^{-1}$
 (D) It cannot be determined from the information given.
33. The temperature-dependence of the rate constants of a series of six related reactions are studied, and the reactions are all found to have the same rate constant at 370 K (their isokinetic temperature). Which statement best describes the relationship between the reactions' Arrhenius pre-factors A and their activation energies E_a ?
- (A) The reactions share a common pre-factor A but differ in their activation energies E_a .
 (B) The reactions share a common activation energy E_a but differ in their pre-factors A .
 (C) As the reactions' pre-factors A increase, their activation energies E_a increase.
 (D) As the reactions' pre-factors A increase, their activation energies E_a decrease.
34. Which properties of the reaction whose reaction coordinate is shown below increase with increasing temperature?



- I. The forward rate constant of the reaction, k_f
 II. The equilibrium constant of the reaction, K_{eq}

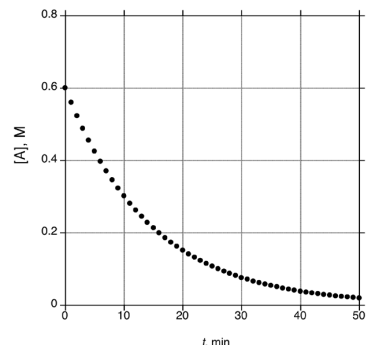
- (A) I only
 (B) II only
 (C) Both I and II
 (D) Neither I nor II

35. The hydrolysis of formamide to formate ion in basic aqueous solution is proposed to proceed by the following steps:



Which of the following statements is necessarily true, given this mechanism?

- (A) The reaction is first-order in HCONH_2 .
 (B) The reaction is first-order in hydroxide ion.
 (C) HCO_2^- is an intermediate in this reaction.
 (D) If the reaction is carried out in deuterium oxide, the formate ion will contain deuterium.
36. An irreversible reaction $A + B \rightarrow C$ is first-order in both A and B. In a solution with $[A]_0 = 0.20 \text{ M}$, it takes 37 s for the concentration of B to decrease from its initial value of 0.010 M to 0.0050 M. How long will it take the concentration of B to decrease from an initial value of 0.010 M to 0.0050 M if the initial concentration of A is 0.010 M?
- (A) 37 s (B) 110 s (C) 740 s (D) 1100 s
37. A reaction $A \rightarrow 2 B$ was monitored over time. What is the average rate of disappearance of A between $t = 0$ and $t = 10$ minutes?



- (A) 0.03 M min^{-1} (B) 0.04 M min^{-1}
 (C) 0.06 M min^{-1} (D) 0.07 M min^{-1}

38. An irreversible reaction $A + B \rightarrow \text{products}$ is monitored in buffered aqueous solution under conditions where $[B] \gg [A]$ by measuring the concentration of A as a function of time. The decay of A is found to be first-order under these conditions, with the half-life for its disappearance varying as shown. What is the rate law for this reaction?

[B]	pH	Half-life for disappearance of A
0.100 M	1.00	120 s
0.200 M	1.00	60 s
0.200 M	1.50	190 s

- (A) $\text{Rate} = k[A][B][H^+]$ (B) $\text{Rate} = k[A]/[B]$
 (C) $\text{Rate} = k[A][H^+]^2/[B]$ (D) $\text{Rate} = k[A][B][H^+]^2$

39. The thermal decomposition of NOCl is a second-order process, and the rate constant k for the disappearance of NOCl at 160 °C is $0.0037 \text{ M}^{-1} \text{ s}^{-1}$. What is the concentration of NOCl, initially at 0.043 M, after 20.0 minutes at 160 °C?

- (A) 0.00051 M (B) 0.036 M
 (C) 0.040 M (D) 0.042 M

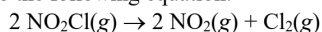
40. Bismuth-212 undergoes both α decay (35.9%) and β decay (64.1%) with an overall half-life of 60.6 minutes. What is its half-life for α decay?

- (A) 21.8 min (B) 38.8 min
 (C) 94.7 min (D) 169 min

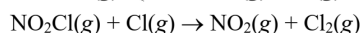
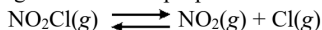
41. A reaction $A \rightleftharpoons B$ is reversible, with $K_{\text{eq}} = 1.00$. The forward reaction is first-order in A. In a reaction initially containing only A at an initial concentration of $[A]_0$, which plot will be linear over the course of the reaction?

- (A) $\ln([A])$ vs. time
 (B) $\ln([A] - 0.5[A]_0)$ vs. time
 (C) $1/[A]$ vs. time
 (D) $1/([A] - 0.5[A]_0)$ vs. time

42. The decomposition of nitryl chloride takes place according to the following equation:



The following mechanism is proposed for this reaction:

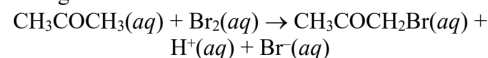


Which rate laws are INCONSISTENT with the proposed mechanism, regardless of the relative rates of the two steps?

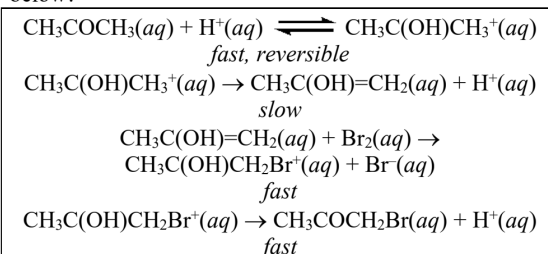
- I. $\text{Rate} = k[\text{NO}_2\text{Cl}]$
 II. $\text{Rate} = k[\text{NO}_2\text{Cl}]^2$

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

43. Acetone is brominated in acidic solution according to the following reaction:



What is the rate law implied by the mechanism given below?



- (A) $\text{Rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+]$
 (B) $\text{Rate} = k[\text{CH}_3\text{COCH}_3]$
 (C) $\text{Rate} = k[\text{CH}_3\text{COCH}_3][\text{Br}_2]$
 (D) $\text{Rate} = \frac{k[\text{CH}_3\text{COCH}_3][\text{Br}_2]}{[\text{H}^+]}$

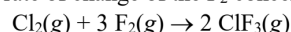
44. A rule of thumb is that a reaction's rate roughly doubles for every 10 °C increase in temperature. What is the activation energy of a reaction whose rate exactly doubles between 25.0 °C and 35.0 °C?

- (A) 52.9 kJ mol⁻¹ (B) 153 kJ mol⁻¹
 (C) 504 kJ mol⁻¹ (D) 523 kJ mol⁻¹

45. At 650 K, β -pinene isomerizes to form either 4-isopropenyl-1-methylcyclohexene or myrcene. The former reaction has a rate constant of 0.22 s^{-1} and the latter a rate constant of 0.13 s^{-1} . What is the overall rate law for the isomerization of β -pinene?

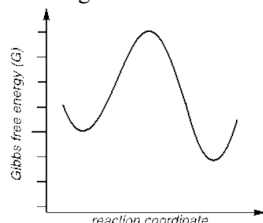
- (A) $\text{Rate} = (0.029 \text{ s}^{-1})[\beta\text{-pinene}]$
 (B) $\text{Rate} = (0.082 \text{ s}^{-1})[\beta\text{-pinene}]$
 (C) $\text{Rate} = (0.17 \text{ s}^{-1})[\beta\text{-pinene}]$
 (D) $\text{Rate} = (0.35 \text{ s}^{-1})[\beta\text{-pinene}]$

46. How is the rate of change of the ClF_3 concentration related to the rate of change of the F_2 concentration?

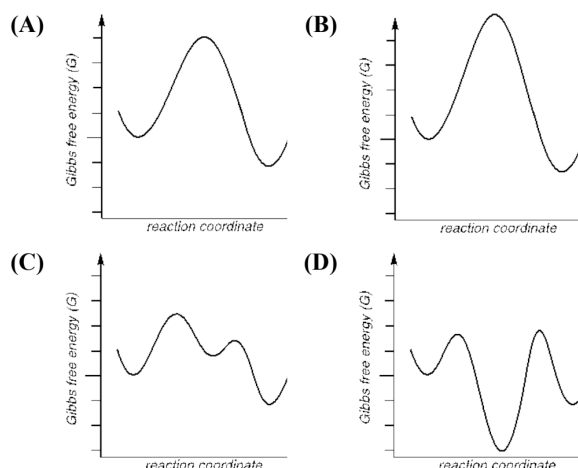


- (A) $\Delta[\text{ClF}_3]/\Delta t = (3/2)(\Delta[\text{F}_2]/\Delta t)$
 (B) $\Delta[\text{ClF}_3]/\Delta t = \Delta[\text{F}_2]/\Delta t$
 (C) $\Delta[\text{ClF}_3]/\Delta t = -(2/3)(\Delta[\text{F}_2]/\Delta t)$
 (D) $\Delta[\text{ClF}_3]/\Delta t = -(3/2)(\Delta[\text{F}_2]/\Delta t)$

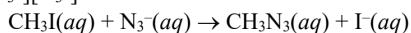
47. A reaction takes place by an uncatalyzed pathway whose reaction coordinate diagram is shown below.



The reaction is accelerated by addition of a small amount of a catalyst. Which reaction coordinate diagram (drawn to the same scale as the one above) best describes the catalyzed reaction?



48. Methyl iodide reacts irreversibly with azide ion with rate $= k[\text{CH}_3\text{I}][\text{N}_3^-]$.



The reaction is carried out with an initial concentration of CH_3I of 0.01 M. Which statement about the reaction is correct?

- (A) The time it takes for $[\text{CH}_3\text{I}]$ to decrease to 0.005 M is independent of $[\text{N}_3^-]$, as long as $[\text{N}_3^-] \gg [\text{CH}_3\text{I}]$.
- (B) If the initial concentrations of azide and CH_3I are equal, then it takes half as long for $[\text{CH}_3\text{I}]$ to decrease to 0.005 M as it does for it to decrease from 0.005 M to 0.0025 M.
- (C) The reaction rate is significantly smaller if excess I^- is added to the solution.
- (D) The reaction cannot take place in a single elementary step.
49. The rate constant k for the reaction shown is $1.63 \times 10^{-6} \text{ M}^{-1} \text{ s}^{-1}$. What is the overall order of the reaction?
- $$2 \text{ICl}(g) + \text{H}_2(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$$
- (A) 0 (B) 1 (C) 2 (D) 3

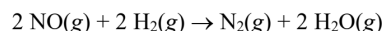
50. The initial rate of the reaction $\text{A} + \text{B} \rightarrow \text{products}$ varies with the initial concentrations of A and B as follows:

[A], M	[B], M	Initial rate, M s^{-1}
0.20	0.30	2.1×10^{-4}
0.40	0.30	4.2×10^{-4}
0.20	0.90	1.9×10^{-3}

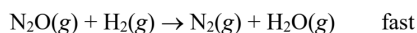
What is the rate law for this reaction?

- (A) $\text{Rate} = k[\text{A}]$ (B) $\text{Rate} = k[\text{A}][\text{B}]$
 (C) $\text{Rate} = k[\text{A}]^2[\text{B}]$ (D) $\text{Rate} = k[\text{A}][\text{B}]^2$
51. Polonium-210 undergoes alpha decay with a half-life of 138 d, forming the stable isotope lead-206. How much time would it take for the activity of a ^{210}Po sample to decrease to 15.0% of its initial value?
- (A) 138 d (B) 182 d (C) 276 d (D) 377 d
52. An irreversible reaction $\text{A} + \text{B} \rightarrow \text{products}$ is studied under conditions where the initial concentrations of A and B are equal. Under these circumstances, a graph of $\ln([\text{A}])$ as a function of time is linear. What is the order in A?
- (A) Zeroth order
 (B) First order
 (C) Second order
 (D) The order in A cannot be determined based on the information given.

53. Nitric oxide reacts with hydrogen according to the following equation:



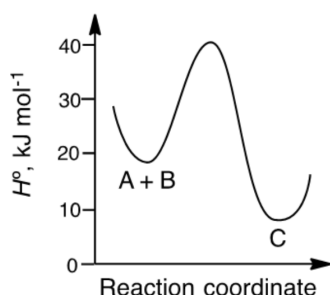
A possible mechanism for this reaction is



What rate law is implied by this mechanism?

- (A) $\text{Rate} = k[\text{NO}][\text{H}_2]$ (B) $\text{Rate} = k[\text{NO}]^2[\text{H}_2]$
 (C) $\text{Rate} = k[\text{NO}][\text{H}_2]^2$ (D) $\text{Rate} = k[\text{NO}]^2[\text{H}_2]^2$

54. The reaction $A + B \rightarrow C$ is represented by the energy diagram shown.



Which statements about this reaction are correct?

- I. The overall reaction is endothermic.
 II. The activation enthalpy of the reaction is 40 kJ mol^{-1} .

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

55. For the reaction $A + B \rightarrow \text{products}$, the rate law is $\text{rate} = k[A]^2[B]$. Which change will cause the greatest decrease in reaction rate?

- (A) Decreasing $[A]$ by a factor of 2
 (B) Decreasing $[B]$ by a factor of 2
 (C) Decreasing both $[A]$ and $[B]$ by a factor of 2
 (D) Decreasing $[B]$ by a factor of 4

56. For a reaction with an activation energy of 65 kJ mol^{-1} , by what percentage is the rate constant decreased if the temperature is decreased from 37°C to 22°C ?

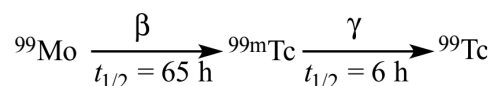
- (A) 13% (B) 27% (C) 51% (D) 72%

57. A plot of $\ln[A]$ as a function of time in an irreversible reaction $A \rightarrow \text{products}$ is linear, with a slope of -0.0175 s^{-1} . What conclusions may be drawn from these observations?

- I. The reaction is first-order in A.
 II. The rate constant for the reaction is 0.0175 s^{-1} .

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

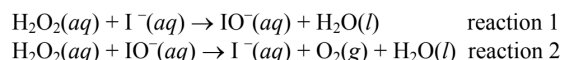
58. A sample containing only the isotope ^{99}Mo undergoes radioactive decay as shown:



Which of the following statements about the relative activity of ^{99}Mo and $^{99\text{m}}\text{Tc}$ in the sample is correct?

- (A) The activity of $^{99\text{m}}\text{Tc}$ exceeds that of the ^{99}Mo after about 20 h.
 (B) The activity of $^{99\text{m}}\text{Tc}$ exceeds that of the ^{99}Mo after about 120 h.
 (C) The activity of $^{99\text{m}}\text{Tc}$ becomes roughly equal to that of the ^{99}Mo after about 20 h.
 (D) The activity of $^{99\text{m}}\text{Tc}$ becomes roughly equal to that of the ^{99}Mo after about 120 h.

59. Hydrogen peroxide disproportionates to water and molecular oxygen in the presence of iodide in neutral solution according to a mechanism consisting of two elementary steps:

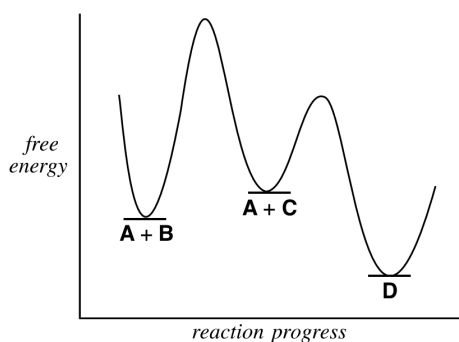


The rate constant for reaction 1 is much larger than the rate constant for reaction 2. Which statement is correct?

- (A) As the reaction proceeds, the predominant form of iodine in solution is $\text{IO}^-(\text{aq})$.
 (B) Adding more iodide to the reaction will not increase the rate of production of O_2 .
 (C) The reaction is zeroth-order in H_2O_2 .
 (D) The reaction will go more slowly at higher O_2 pressures.

Kinetics

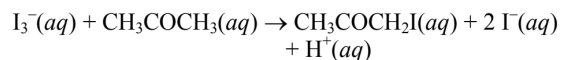
60. Which of the following are true about the overall reaction $A + B \rightarrow D$ illustrated in the diagram?



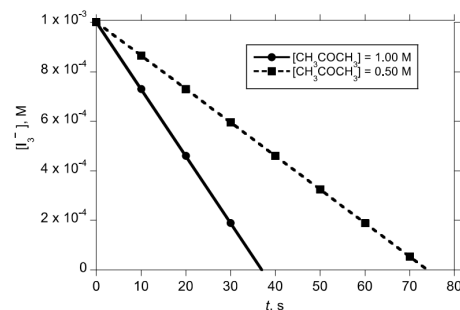
- I. The reaction displays second-order kinetics.
II. The reaction has two intermediates.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
61. Cyclopropane isomerizes to propene in an irreversible, first-order reaction. At 700 K, a sample of 22.0 mm Hg of cyclopropane is introduced into a reaction vessel. After 1.0 min, the partial pressure of the product, propene, was found to be 17.5 mm Hg. What is the rate constant for the isomerization at this temperature?
- (A) $3.8 \times 10^{-3} \text{ s}^{-1}$ (B) $2.6 \times 10^{-2} \text{ s}^{-1}$
(C) 0.23 s^{-1} (D) 1.6 s^{-1}
62. Which statements about the collision theory of reactions are correct?
- I. Molecules must have the correct spatial orientations for collisions to lead to reactions.
II. Only collisions with an energy greater than a certain threshold lead to reactions.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

63. Triiodide reacts with acetone in acidic aqueous solution:

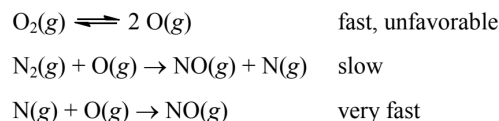


The reaction is carried out twice, both times with $[\text{I}_3^-]_0 = 1.0 \times 10^{-3} \text{ M}$ and $[\text{H}^+]_0 = 1.00 \text{ M}$, but with two different concentrations of acetone, and the concentration of triiodide is found to vary with time as shown:



What are the orders of triiodide and acetone in the reaction?

- (A) First-order in I_3^- , first-order in CH_3COCH_3
(B) Zeroth-order in I_3^- , first-order in CH_3COCH_3
(C) First-order in I_3^- ; the order in CH_3COCH_3 cannot be determined from the available data
(D) First-order in CH_3COCH_3 ; the order in I_3^- cannot be determined from the available data
64. Nitric oxide is formed at high temperature in the presence of oxygen and nitrogen. A proposed mechanism for its formation is shown:



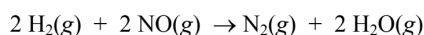
What rate law is predicted by this mechanism?

- (A) $\text{Rate} = k[\text{O}_2]$ (B) $\text{Rate} = k[\text{N}_2][\text{O}_2]$
(C) $\text{Rate} = k[\text{N}_2][\text{O}_2]^{1/2}$ (D) $\text{Rate} = k[\text{N}_2][\text{O}_2]^2$
65. At 400.0 K, the rate constant of a reaction is $3.2 \times 10^{-2} \text{ s}^{-1}$. The activation energy of the reaction is 41.0 kJ mol^{-1} . What is the rate constant at 410.0 K?
- (A) $2.4 \times 10^{-2} \text{ s}^{-1}$ (B) $3.2 \times 10^{-2} \text{ s}^{-1}$
(C) $4.3 \times 10^{-2} \text{ s}^{-1}$ (D) $3.9 \times 10^{-1} \text{ s}^{-1}$

Kinetics

66. In a reaction between two substances A and B, the rate is found to be proportional to the concentration of A at all concentrations studied. At low concentrations of B, the rate of the reaction is found to be proportional to [B], but to level off at high concentrations of B. What is a reasonable explanation for the observed dependence of rate on [B]?
- (A) B binds to A prior to the rate-determining step.
 (B) The reaction has an order of -1 in B.
 (C) The reaction involves a 1:2 stoichiometry of A:B.
 (D) The reaction is catalyzed by one of its products.

67. For the reaction

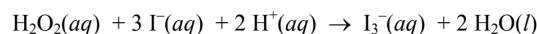


the rate law is $\text{rate} = k[\text{H}_2][\text{NO}]^2$. At a given temperature, what is the effect on the reaction rate if the concentration of H_2 is doubled and the concentration of NO is halved?

- (A) The reaction rate is halved.
 (B) The reaction rate is unchanged.
 (C) The reaction rate is doubled.
 (D) The reaction rate increases eightfold.

68. A substance X decomposes in a second-order reaction. A solution that is initially 1.00 M in X requires 0.50 h for its concentration to decrease to 0.50 M. How much time will it take for a solution of X to decrease in concentration from 1.00 M to 0.25 M?
- (A) 0.50 h (B) 1.0 h (C) 1.5 h (D) 2.0 h

69. The reaction



has a rate law of $\text{rate} = k[\text{H}_2\text{O}_2][\text{I}^-]$. What is the order of the reaction with respect to H^+ , and what is the overall order of the reaction?

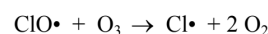
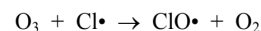
- (A) 0th order in H^+ , 2nd order overall
 (B) 1st order in H^+ , 2nd order overall
 (C) 1st order in H^+ , 3rd order overall
 (D) 2nd order in H^+ , 6th order overall

70. Which of the following are reasons why reaction rates increase as temperature increases?

- I. Collisions are more frequent between molecules at higher temperatures.
 II. A greater fraction of collisions have sufficient energy to exceed E_a at higher temperatures.
 III. Reactant concentrations are higher at higher temperatures.

- (A) I only (B) II only
 (C) I and II (D) I, II, and III

71. What is the effect of adding a catalyst on the rate of a reversible reaction in the forward and the reverse direction?
- (A) It has no effect on the rate in either direction.
 (B) Both rates increase by the same factor.
 (C) The rate in the forward direction increases by a greater factor than the rate in the reverse direction.
 (D) The rate in the reverse direction increases by a greater factor than the rate in the forward direction.
72. Consider the proposed mechanism for the destruction of ozone in the stratosphere:



Which of the statements about this mechanism is correct?

- (A) $\text{Cl}\cdot$ is a catalyst.
 (B) O_2 is an intermediate.
 (C) Equal amounts of $\text{Cl}\cdot$ and $\text{ClO}\cdot$ are present.
 (D) The number of moles of O_2 produced equals the number of moles of O_3 consumed.

73. When the kinetics of the reaction, $2 \text{A} + 2 \text{B} \rightarrow \text{C}$, were studied using the method of initial rates, the data in the table below were obtained.

Trial	$[\text{A}]_0$ in $\text{mol}\cdot\text{L}^{-1}$	$[\text{B}]_0$ in $\text{mol}\cdot\text{L}^{-1}$	Initial Rate of Formation of C in $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
1	0.060	0.040	3.6×10^{-4}
2	0.060	0.080	7.2×10^{-4}
3	0.030	0.12	5.4×10^{-4}

What is the rate law for the reaction?

- (A) $\text{Rate} = k[\text{A}][\text{B}]$ (B) $\text{Rate} = k[\text{A}]^2[\text{B}]$
 (C) $\text{Rate} = k[\text{A}][\text{B}]^2$ (D) $\text{Rate} = k[\text{A}]^2[\text{B}]^2$

74. $2 \text{NO}_2(\text{g}) \rightarrow 2 \text{NO}(\text{g}) + \text{O}_2(\text{g})$

For the reaction above, a straight line for which plot indicates a second order reaction?

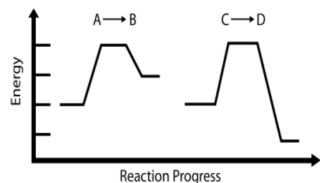
- (A) $[\text{NO}_2]$ vs. time (B) $[\text{NO}_2]^2$ vs. time
 (C) $1/[\text{NO}_2]$ vs. time (D) $1/[\text{NO}_2]^2$ vs. time

75. What are the units for the rate constant of a zero-order reaction?

- (A) time (B) time^{-1}
 (C) $\text{M}\cdot\text{time}$ (D) $\text{M}\cdot\text{time}^{-1}$

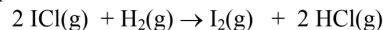
Kinetics

76. From the energy profiles for the two unimolecular reactions below, how will the rates of the two reactions compare if the temperature of each reaction is increased from 25 °C to 75 °C?



- (A) The rate of reaction $A \rightarrow B$ will increase more than the rate of reaction $C \rightarrow D$.
- (B) The rate of reaction $C \rightarrow D$ will increase more than the rate of reaction $A \rightarrow B$.
- (C) The rates of the two reactions will increase by the same amount.
- (D) The rate of reaction $A \rightarrow B$ will increase but the rate of reaction $C \rightarrow D$ will decrease.
77. The half-life of a first-order reaction is 1.5 hours. How much time is needed for 94% of the reactant to change to product?
- (A) 0.13 hours (B) 6.1 hours
(C) 2.3 hours (D) 36 hours
78. For the reaction, $2A + B \rightarrow C + D$, the rate law is $\text{rate} = k[B]$. Which of the following mechanisms would be consistent with this information?
- (A) $A + B \rightarrow M$ (slow) (B) $A + A \rightarrow M$ (fast)
 $A + M \rightarrow C + D$ (fast) $B + M \rightarrow C + D$ (slow)
- (C) $B \rightarrow M$ (slow) (D) $B \rightarrow M$ (fast)
 $M + A \rightarrow N$ (fast) $M + A \rightarrow N$ (slow)
 $N + A \rightarrow C + D$ (fast) $N + A \rightarrow C + D$ (fast)
79. Ammonia reacts with oxygen according to the equation:
 $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$
 In an experiment in which the rate of change of nitric oxide is found to be $1.10\text{ M}\cdot\text{min}^{-1}$, what is the rate of change of oxygen gas?
- (A) $-1.38\text{ M}\cdot\text{min}^{-1}$ (B) $-0.880\text{ M}\cdot\text{min}^{-1}$
(C) $-0.275\text{ M}\cdot\text{min}^{-1}$ (D) $-0.220\text{ M}\cdot\text{min}^{-1}$
80. A first-order reaction has a rate constant of $k = 0.320\text{ min}^{-1}$. For an initial reactant concentration of 1.22 M, how long does it take for its concentration to fall to 0.150 M?
- (A) 0.671 min (B) 2.60 min
(C) 6.55 min (D) 25.4 min

81. Iodine monochloride reacts with hydrogen according to the equation:



The rate data below have been obtained.

$[\text{ICl}]$, M	$[\text{H}_2]$, M	Initial Rate, $\text{M}\cdot\text{s}^{-1}$
0.10	0.10	0.030
0.20	0.10	0.060
0.10	0.050	0.015

What is the rate law for the reaction?

- (A) $\text{Rate} = k[\text{ICl}][\text{H}_2]$ (B) $\text{Rate} = k[\text{ICl}]^2[\text{H}_2]$
(C) $\text{Rate} = k[\text{ICl}][\text{H}_2]^{1/2}$ (D) $\text{Rate} = k[\text{ICl}]^2[\text{H}_2]^{1/2}$
82. Which graph is linear for a reaction that is second order in $[A]$?
- (A) $[A]$ vs time (B) $\ln[A]$ vs time
(C) $[A]^2$ vs time (D) $1/[A]$ vs time
83. The reaction of nitrogen dioxide with carbon monoxide
 $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{NO}(\text{g})$
 has been studied and the following mechanism has been proposed:
 $\text{NO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{NO}_3(\text{g}) + \text{NO}(\text{g})$ slow
 $\text{NO}_3(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{CO}_2(\text{g})$ fast
 What rate law corresponds to this mechanism?
- (A) $\text{Rate} = k[\text{NO}_2]$ (B) $\text{Rate} = k[\text{NO}_2][\text{CO}]$
(C) $\text{Rate} = k[\text{NO}_2]^2$ (D) $\text{Rate} = k[\text{NO}_2]^2[\text{CO}]$
84. A reaction has a rate constant $k = 8.54 \times 10^{-4}\text{ M}^{-1}\cdot\text{s}^{-1}$ at 45 °C and an activation energy, $E_a = 90.8\text{ kJ}$. What is the value of k at 25 °C?
- (A) $4.46 \times 10^{-5}\text{ M}^{-1}\cdot\text{s}^{-1}$ (B) $8.54 \times 10^{-5}\text{ M}^{-1}\cdot\text{s}^{-1}$
(C) $8.52 \times 10^{-4}\text{ M}^{-1}\cdot\text{s}^{-1}$ (D) $8.54 \times 10^{-3}\text{ M}^{-1}\cdot\text{s}^{-1}$
85. Which elementary reaction characteristic(s) change(s) significantly for a 10 °C temperature increase for a reaction carried out near room temperature?
- I fraction of molecules with required E_a
 II fraction of molecules with correct orientation
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Kinetics

86. Hemoglobin (Hb) reacts with carbon monoxide according to the equation $4\text{Hb} + 3\text{CO} \rightarrow \text{Hb}_4(\text{CO})_3$. What is the rate law for this reaction at 20 °C?

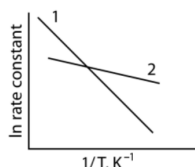
Trial	[Hb], M	[CO], M	Initial rate of disappearance of Hb $\text{M}\cdot\text{s}^{-1}$
1	1.50×10^{-6}	1.00×10^{-6}	9.20×10^{-7}
2	3.00×10^{-6}	1.00×10^{-6}	1.84×10^{-6}
3	3.00×10^{-6}	3.00×10^{-6}	5.52×10^{-6}

- (A) Rate = $k[\text{Hb}][\text{CO}]$ (B) Rate = $k[\text{Hb}][\text{CO}]^2$
 (C) Rate = $k[\text{Hb}]^2[\text{CO}]$ (D) Rate = $k[\text{Hb}][\text{CO}]^3$

87. What is the first-order rate constant for a reaction that is 36.5% complete in 0.0200 seconds?

- (A) 50.4 s^{-1} (B) 27.7 s^{-1} (C) 22.7 s^{-1} (D) 9.86 s^{-1}

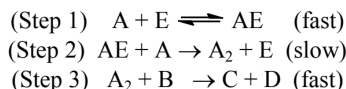
88.



The diagram above depicts the temperature behavior of the rate constant, k , for two reactions, 1 and 2. Which statement about the k values at low temperatures and the activation energies, E_a , for these reactions is correct?

	k values at low T	E_a values
(A)	$k_{\text{rxn 1}} < k_{\text{rxn 2}}$	$E_{a \text{ rxn 1}} < E_{a \text{ rxn 2}}$
(B)	$k_{\text{rxn 1}} < k_{\text{rxn 2}}$	$E_{a \text{ rxn 1}} > E_{a \text{ rxn 2}}$
(C)	$k_{\text{rxn 1}} > k_{\text{rxn 2}}$	$E_{a \text{ rxn 1}} < E_{a \text{ rxn 2}}$
(D)	$k_{\text{rxn 1}} > k_{\text{rxn 2}}$	$E_{a \text{ rxn 1}} > E_{a \text{ rxn 2}}$

89. The hypothetical reaction $2\text{A} + \text{B} \rightarrow \text{C} + \text{D}$ is catalyzed by E as indicated in the possible mechanism below.



Which rate law best agrees with this mechanism?

- (A) Rate = $k[\text{A}][\text{B}]$ (B) Rate = $k[\text{A}][\text{E}]$
 (C) Rate = $k[\text{A}]^2[\text{E}]$ (D) Rate = $k[\text{A}]^2[\text{B}]$

90. A rigid container holds an equal number of moles of N_2 and H_2 gas at a total pressure of 10.0 atm. The gases react according to the equation, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. If the total pressure of the gas decreases at a rate of $0.20 \text{ atm}\cdot\text{s}^{-1}$, what is the rate of change of the partial pressure of N_2 in the container?

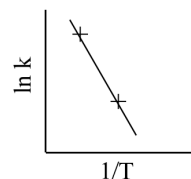
- (A) decreases at $0.40 \text{ atm}\cdot\text{s}^{-1}$
 (B) decreases at $0.30 \text{ atm}\cdot\text{s}^{-1}$
 (C) decreases at $0.20 \text{ atm}\cdot\text{s}^{-1}$
 (D) decreases at $0.10 \text{ atm}\cdot\text{s}^{-1}$

91. A reaction is found to have the rate law: Rate = $k[\text{A}]^2[\text{B}]$. Which set of data is consistent with this finding?

	[A], $\text{mol}\cdot\text{L}^{-1}$	[B], $\text{mol}\cdot\text{L}^{-1}$	Rate, $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
Given	0.010	0.010	1.2×10^{-4}
A	0.010	0.020	2.4×10^{-4}
B	0.020	0.010	6.0×10^{-4}
C	0.020	0.020	4.8×10^{-4}
D	0.010	0.020	4.8×10^{-4}

- (A) A (B) B (C) C (D) D

92. The value of the specific rate constant, k , for a reaction is determined at two different temperatures and plotted in the accompanying graph.



What is the relationship between the slope of the graph and the activation energy, E_a ?

- (A) slope = E_a (B) slope = $-E_a$
 (C) slope = $-E_a/R$ (D) slope = $E_a \times R$

93. Consider the reaction, $2\text{NO}(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{NOCl}$. Which of the following is/are required for a successful reaction between NO and Cl_2 molecules?

- I Proper orientation
 II NO/ Cl_2 Ratio of 2 to 1
 III Sufficient collision energy

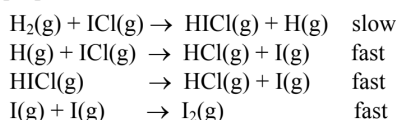
- (A) II only (B) I and III only
 (C) II and III only (D) I, II, and III

Kinetics

94. When sucrose, $C_{12}H_{22}O_{11}$, is dissolved in H_2O in the presence of an acid catalyst it reacts according to the equation $C_{12}H_{22}O_{11} + H_2O \rightarrow 2 C_6H_{12}O_6$ with a rate law of $Rate = k[C_{12}H_{22}O_{11}]$. If 3.00 g of sucrose decreases to 2.70 g in 2.50 hours in the presence of a certain concentration of an acid catalyst, what is the half-life for this reaction under these same conditions?

(A) 12.5 hours (B) 16.4 hours
(C) 23.7 hours (D) 37.9 hours

95. For the reaction $H_2(g) + 2 ICl(g) \rightarrow 2 HCl(g) + I_2(g)$ one proposed mechanism is



Intermediates in this reaction include which of the following?

(A) $HICl$ only (B) I only
(C) $HICl$ and H only (D) $HICl$, H , and I

96. The rate of decomposition of a certain compound in solution is first order. If the concentration of the compound is doubled, what happens to the reaction's half-life?

(A) It doubles
(B) It decreases to $\frac{1}{2}$ of the original value
(C) It decreases to $\frac{1}{4}$ of the original value
(D) It remains the same

97. Consider the reaction: $2 ICl(g) + H_2(g) \rightarrow 2 HCl(g) + I_2(g)$. At a certain temperature the rate constant is found to be $1.63 \times 10^{-6} \text{ L/mol}\cdot\text{s}$. What is the overall order of the reaction?

(A) zero (B) first (C) second (D) third

98. For the reaction: $N_2O_4(g) \rightarrow 2 NO_2(g)$ the number of moles of $N_2O_4(g)$ is

Time, min	0	5	10
Moles $N_2O_4(g)$	0.200	0.170	0.140

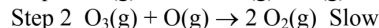
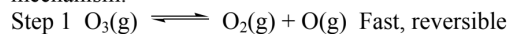
What is the number of moles of $NO_2(g)$ at $t = 10 \text{ min}$? (Assume moles of $NO_2(g) = 0$ at $t = 0$.)

(A) 0.280 (B) 0.120 (C) 0.110 (D) 0.060

99. A compound decomposes with a first-order rate constant of 0.00854 s^{-1} . Calculate the concentration after 5.0 minutes for an initial concentration of 1.2 M.

(A) 0.010 M (B) 0.093 M
(C) 0.92 M (D) 1.1 M

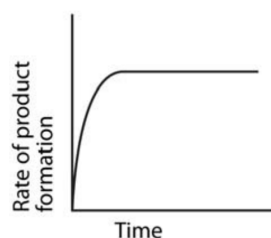
100. Ozone in the earth's atmosphere decomposes according to the equation: $2 O_3(g) \rightarrow 3 O_2(g)$. This reaction is thought to occur via the two-step mechanism:



What rate law is consistent with this mechanism?

(A) $-\Delta[O_3]/\Delta t = k[O_3]$
(B) $-\Delta[O_3]/\Delta t = k[O_3]^2$
(C) $-\Delta[O_3]/\Delta t = k[O_3]^2/[O_2]$
(D) $-\Delta[O_3]/\Delta t = k[O_3]^2/[O_2]^3$

101.



The rates of many substrate reactions catalyzed by enzymes vary with time as shown. Which factor(s) best account(s) for the constant reaction rate after a certain time?

I The enzyme's active sites are filled.
II The amount of substrate is constant.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

102. What is the rate law for a reaction, $A + 2B \rightarrow C$, based on the rate data?

$[A]_0, \text{mol}\cdot\text{L}^{-1}$	$[B]_0, \text{mol}\cdot\text{L}^{-1}$	Initial Rate, $\text{mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$
0.20	0.10	300
0.40	0.30	3600
0.80	0.30	14400

(A) $Rate = k[A][B]$ (B) $Rate = k[A][B]^2$
(C) $Rate = k[A]^2[B]$ (D) $Rate = k[A]^2[B]^3$

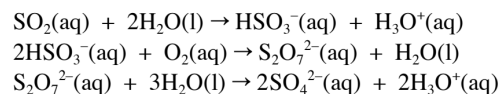
103. For the irreversible reaction; $A \rightarrow B$, which graph gives a straight line for a reaction that is second order in A?

(A) $[A]$ vs. time (B) $1/[A]$ vs. time
(C) $[A]^2$ vs. time (D) $1/[A]^2$ vs. time

104. A sample of a radioactive element that contains 1.0×10^3 nuclei decays to 6.2×10^1 nuclei in 10. minutes. What is its specific decay constant?

(A) 6.2 min^{-1} (B) 1.2 min^{-1}
(C) 0.28 min^{-1} (D) 0.062 min^{-1}

105. The oxidation of SO_2 to H_2SO_4 in acid rain is thought to occur by the following mechanism.



Which species in this mechanism can be given the following designations?

	reactant	catalyst	intermediate
(A)	$\text{SO}_2(\text{aq})$	$\text{H}_2\text{O}(\text{l})$	$\text{HSO}_3^-(\text{aq}), \text{H}_3\text{O}^+(\text{aq})$
(B)	$\text{SO}_2(\text{aq})$	$\text{HSO}_3^-(\text{aq})$	$\text{S}_2\text{O}_7^{2-}(\text{aq})$
(C)	$\text{SO}_2(\text{aq}), \text{H}_2\text{O}(\text{l})$	$\text{S}_2\text{O}_7^{2-}(\text{aq})$	$\text{HSO}_3^-(\text{aq})$
(D)	$\text{SO}_2(\text{aq}), \text{H}_2\text{O}(\text{l})$	none	$\text{HSO}_3^-(\text{aq}), \text{S}_2\text{O}_7^{2-}(\text{aq})$

106. For the reaction; $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$, the rate law is $\text{Rate} = k[\text{H}_2][\text{I}_2]$. Which of the mechanisms given is consistent with this rate law?

I.	II.
$\text{I}_2 \rightleftharpoons \text{I} + \text{I}$ (fast)	$\text{I}_2 \rightleftharpoons \text{I} + \text{I}$ (slow)
$\text{H}_2 + \text{I} + \text{I} \rightarrow 2\text{HI}$ (slow)	$\text{H}_2 + \text{I} + \text{I} \rightarrow 2\text{HI}$ (fast)

- (A) I only
(B) II only
(C) Either I or II
(D) Neither I nor II

107. The rate of a reaction at 100°C is four times the rate at 50°C . What is its activation energy?

- (A) $1152 \text{ kJ}\cdot\text{mol}^{-1}$
(B) $80.1 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $54.0 \text{ kJ}\cdot\text{mol}^{-1}$
(D) $27.8 \text{ kJ}\cdot\text{mol}^{-1}$

108. A student analyzed the data from a zero order reaction and obtained the graph shown. What labels should be attached to the X and Y axes, respectively?



- (A) time, concentration
(B) time, $1 / \text{concentration}$
(C) time, $\ln(\text{concentration})$
(D) $1/\text{time}$, concentration

109. Under certain conditions the reaction of CO with NO_2 to give CO_2 and NO results in the rate law:

$$\text{rate} = k[\text{CO}][\text{NO}_2]$$

What are the units for the rate constant, k ?

- (A) $\text{mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$
(B) $\text{L}\cdot\text{mol}^{-1}\cdot\text{min}^{-1}$
(C) $\text{mol}^2\cdot\text{L}^{-2}\cdot\text{min}^{-1}$
(D) $\text{L}^2\cdot\text{mol}^{-2}\cdot\text{min}^{-1}$

110. For the reaction: $\text{X} + \text{Y} \rightarrow \text{Z}$, initial rate data are given in the table.

$[\text{X}] / \text{M}$	$[\text{Y}] / \text{M}$	Rate / $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
0.10	0.10	0.020
0.10	0.20	0.080
0.30	0.30	0.540

What is the rate law for this reaction?

- (A) $\text{Rate} = k[\text{X}]^2$
(B) $\text{Rate} = k[\text{Y}]^2$
(C) $\text{Rate} = k[\text{X}][\text{Y}]$
(D) $\text{Rate} = k[\text{X}][\text{Y}]^2$

111. The rate of the reaction of chlorine gas with a liquid hydrocarbon can be increased by all of the changes except one. Which change will be ineffective?

- (A) Use UV light to dissociate the Cl_2 .
(B) Increase temperature at constant pressure.
(C) Divide the liquid into small droplets.
(D) Double the pressure by adding He gas.

112. One proposed mechanism of the reaction of HBr with O_2 is given here.

$\text{HBr} + \text{O}_2 \rightarrow \text{HOBr}$	(slow)
$\text{HOBr} + \text{HBr} \rightarrow 2\text{H}_2\text{O}$	(fast)
$\text{HOBr} + \text{HBr} \rightarrow \text{H}_2\text{O} + \text{Br}_2$	(fast)

What is the equation for the overall reaction?

- (A) $\text{HBr} + \text{O}_2 \rightarrow \text{HOBr}$
(B) $2\text{HBr} + \text{O}_2 \rightarrow \text{Br}_2 + \text{H}_2\text{O}_2$
(C) $4\text{HBr} + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + 2\text{Br}_2$
(D) $2\text{HOBr} \rightarrow 2\text{H}_2\text{O} + \text{Br}_2$

113. For the reaction; $\text{A} \rightarrow \text{B}$, the rate law is $\text{rate} = k[\text{A}]$. If the reaction is 40.0% complete after 50.0 minutes, what is the value of the rate constant, k ?

- (A) $8.00 \times 10^{-3} \text{ min}^{-1}$
(B) $1.02 \times 10^{-2} \text{ min}^{-1}$
(C) $1.39 \times 10^{-2} \text{ min}^{-1}$
(D) $1.83 \times 10^{-2} \text{ min}^{-1}$

114. Which of these factors affect the value of the rate constant for a reaction?

- I. temperature
II. reactant concentration
III. use of a catalyst

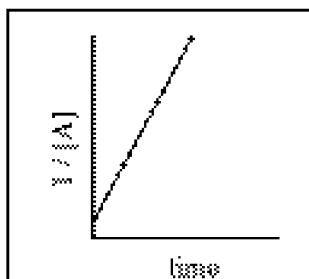
- (A) I only
(B) II only
(C) I and III only
(D) I, II and III

115. Which is the correct exponential form of the Arrhenius equation?

- (A) $E_a = Ae^{-k/RT}$
(B) $E_a = Ae^{k/RT}$
(C) $k = Ae^{-RT/E_a}$
(D) $k = Ae^{-E_a/RT}$

Kinetics

116. For the reaction $A \rightarrow B$, what is the order with respect to A that gives this graph?



(A) zero (B) first (C) second (D) third

117. For the reaction $A \rightarrow B$ that is first-order in A, the rate constant is $2.08 \times 10^{-2} \text{ s}^{-1}$. How long would it take for [A] to change from 0.100 M to 0.0450 M?

(A) 0.0166 s (B) 16.7 s (C) 38.4 s (D) 107 s

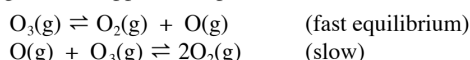
118. These data were obtained for the reaction: $X + Y \rightarrow Z$.

X (M)	Y (M)	Rate: $\Delta Z / \Delta t / \text{M} \cdot \text{min}^{-1}$
1.00	1.00	2.36×10^{-4}
2.00	2.00	1.89×10^{-3}
2.00	4.00	3.78×10^{-3}

What is the rate law?

(A) Rate = $k[X][Y]$ (B) Rate = $k[X]^2[Y]$
(C) Rate = $k[X][Y]^2$ (D) Rate = $k[X]^2[Y]^2$

119. A possible mechanism for the conversion of ozone to oxygen in the upper atmosphere is



Which rate law is consistent with this mechanism?

(A) Rate = $k[\text{O}_3]$ (B) Rate = $k[\text{O}_3]^2$
(C) Rate = $k[\text{O}_3][\text{O}]$ (D) Rate = $k[\text{O}_3]^2[\text{O}_2]^{-1}$

120. How can the rate of reaction at a specific time be determined from a graph of concentration against time?
- (A) concentration at that time divided by the time
(B) logarithm of the concentration divided by the time
(C) absolute value of the slope of the graph at that time
(D) logarithm of the slope divided by the time

121. The rate constant for the radioactive decay of C-11 is 0.0341 min^{-1} . How long will it take for a sample of C-11 to decrease to 1/4 of its original activity?

(A) 20.3 min (B) 29.3 min
(C) 40.6 min (D) 58.6 min

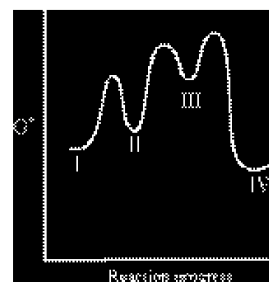
122. If a reaction $A \rightarrow B$ has the rate law $k[A]^2$, which graph produces a straight line?

(A) $1/[A]$ vs time (B) $\ln[A]$ vs time
(C) $[A]^2$ vs time (D) $1/\ln[A]$ vs time

123. Two unimolecular reactions, I and II, have the same rate constant at 25 °C but E_a for reaction I is larger than E_a for reaction II. Which statement about these two reactions is correct?

(A) $k_{\text{reaction I}}$ is the same as $k_{\text{reaction II}}$ at all temperatures.
(B) $k_{\text{reaction I}}$ is larger than $k_{\text{reaction II}}$ at lower temperatures but smaller at higher temperatures.
(C) $k_{\text{reaction I}}$ is smaller than $k_{\text{reaction II}}$ at lower temperatures but larger at higher temperatures.
(D) $k_{\text{reaction I}}$ is larger than $k_{\text{reaction II}}$ at temperatures both lower and higher than 25 °C.

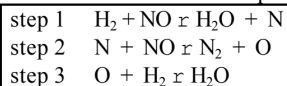
124. According to the reaction profile given, which reaction step is rate-determining in the forward direction?



(A) I $\rightarrow$ II (B) II $\rightarrow$ III
(C) III $\rightarrow$ IV (D) III $\rightarrow$ II

125. For the reaction;
 $2\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$,
rate = $k[\text{H}_2][\text{NO}]^2$.

This mechanism has been proposed:



Which statement about this rate law and mechanism is correct?

(A) This mechanism is consistent with the rate law IF step 1 is the rate determining step.
(B) This mechanism is consistent with the rate law IF step 2 is the rate determining step.
(C) This mechanism is consistent with the rate law IF step 3 is the rate determining step.
(D) This mechanism can not be consistent with the rate law, regardless of which step is rate-determining.

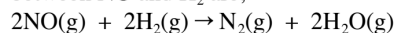
126. For the reaction $A \rightarrow B$ which is first order in A, which of the following change as the concentration of A changes?

I.	rate
II.	rate constant
III.	Half-life

(A) I only (B) III only
(C) II and III only (D) I, II and III

Kinetics

127. The equation and rate law for the gas phase reaction between NO and H₂ are;



$$\text{Rate} = k[\text{NO}]^2[\text{H}_2]$$

What are the units of k if time is in seconds and the concentration is in moles per liter?

- (A) $\text{L} \cdot \text{s} \cdot \text{mol}^{-1}$ (B) $\text{L}^2 \cdot \text{mol}^{-2} \cdot \text{s}^{-1}$
(C) $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$ (D) $\text{mol}^2 \cdot \text{L}^{-2} \cdot \text{s}^{-1}$

128. At a given temperature a first-order reaction has a rate constant of $3.33 \times 10^{-3} \text{ s}^{-1}$. How much time is required for the reaction to be 75% complete?

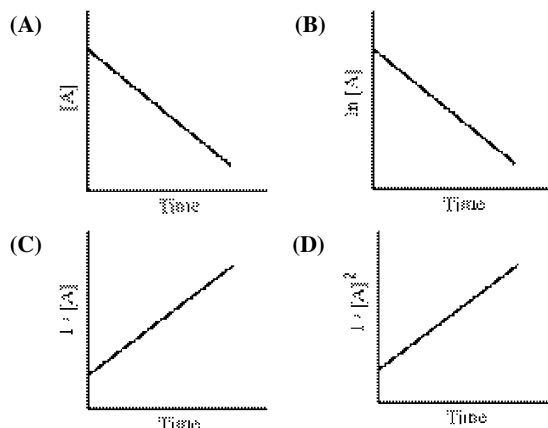
- (A) 100 s (B) 210 s (C) 420 s (D) 630 s

129. Most reactions occur more rapidly at high temperatures than at low temperatures. This is consistent with an increase in which property at higher temperatures?

- I. activation energy
II. collision energy
III. rate constant

- (A) I only (B) II only
(C) I and III only (D) II and III only

130. Which graph is diagnostic of an irreversible second order reaction $\text{A} \rightarrow \text{B}$?



131. The reaction; $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{N}_2(\text{g})$ obeys the rate equation $\text{Rate} = k[\text{NO}]^2[\text{H}_2]$. This mechanism has been proposed:

1. $2\text{NO}(\text{g}) \rightarrow \text{N}_2\text{O}_2(\text{g})$
2. $\text{N}_2\text{O}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow 2\text{HON}(\text{g})$
3. $\text{HON}(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) + \text{HN}(\text{g})$
4. $\text{HN}(\text{g}) + \text{HON}(\text{g}) \rightarrow \text{N}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$

Which step of the mechanism is the rate-determining step?

- (A) step 1 (B) step 2
(C) step 3 (D) step 4

132. The rate constant for a reaction is affected by which factors?

- I increase in temperature
II concentration of the reactants
III presence of a catalyst

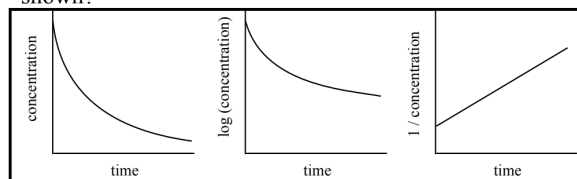
- (A) I and II only (B) I and III only
(C) II and III only (D) I, II and III

133. The rate data given were obtained for the reaction, $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$. What is the rate law for this reaction?

NO pressure (atm)	H ₂ pressure (atm)	Rate (atm·sec ⁻¹)
0.375	0.500	6.43×10^{-4}
0.375	0.250	3.15×10^{-4}
0.188	0.500	1.56×10^{-4}

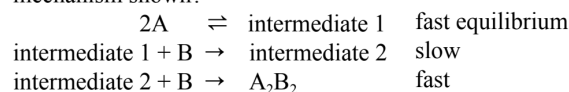
- (A) $\text{Rate} = k P_{\text{NO}}$ (B) $\text{Rate} = k P_{\text{NO}}^2$
(C) $\text{Rate} = k P_{\text{NO}} P_{\text{H}_2}^2$ (D) $\text{Rate} = k P_{\text{NO}}^2 P_{\text{H}_2}$

134. What is the order of a reaction that produces the graphs shown?



- (A) zero order (B) first order
(C) second order (D) some other order

135. What is the rate law for the hypothetical reaction with the mechanism shown?



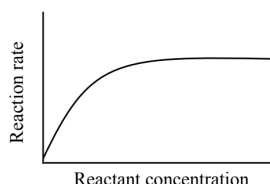
- (A) $\text{Rate} = k[\text{A}]^2$ (B) $\text{Rate} = [\text{B}]^2$
(C) $\text{Rate} = k[\text{A}][\text{B}]$ (D) $\text{Rate} = k[\text{A}]^2[\text{B}]$

136. According to the Arrhenius equation: $k = \text{Ae}^{-E_a/RT}$, a plot of $\ln k$ against $1/T$ yields

- (A) E_a as the slope and A as the intercept
(B) E_a/R as the slope and A as the intercept
(C) E_a/R as the slope and $\ln A$ as the intercept
(D) $-E_a/R$ as the slope and $\ln A$ as the intercept

Kinetics

137. Curves with the shape shown are often observed for reactions involving catalysts. The level portion of the curve is best attributed to the fact that



- (A) product is no longer being formed.
 (B) the reaction has reached equilibrium.
 (C) all the catalytic sites are occupied.
 (D) all the reactant has been consumed.
138. Which statement about second order reactions is correct?
 (A) Second order reactions require different reactants.
 (B) Second order reactions are faster than first order reactions.
 (C) Second order reactions are unaffected by changes in temperature.
 (D) The half-life of a second order reaction depends on the initial reactant concentration.
139. A first order reaction has a rate constant of 0.0541 s^{-1} at 25°C . What is the half-life for this reaction?
 (A) 18.5 s (B) 12.8 s
 (C) 0.0781 s (D) 0.0375 s
140. The reaction between NO and I_2 is second-order in NO and first-order in I_2 . What change occurs in the rate of the reaction if the concentration of each reactant is tripled?
 (A) 3-fold increase (B) 6-fold increase
 (C) 18-fold increase (D) 27-fold increase

141. For the rate equation,

$$\text{Rate} = k[\text{A}][\text{B}]^2,$$

what are the units for the rate constant, k , if the rate is given in $\text{mol}\cdot\text{L}^{-1}\cdot\text{sec}^{-1}$?

- (A) $\text{L}\cdot\text{mol}\cdot\text{sec}$ (B) $\text{L}\cdot\text{mol}^{-1}\cdot\text{sec}^{-1}$
 (C) $\text{L}^2\cdot\text{mol}^{-2}\cdot\text{sec}^{-1}$ (D) $\text{L}^3\cdot\text{mol}^{-3}\cdot\text{sec}^{-2}$
142. For the reaction
 $2\text{A} + 2\text{B} \rightarrow \text{Product}$
 the rate law is $\text{Rate} = k[\text{A}][\text{B}]^2$. Which mechanism is consistent with this information?
 (A) $\text{B} + \text{B} \rightleftharpoons \text{C}$ (B) $\text{A} + \text{B} \rightarrow \text{C}$ (slow)
 $\text{C} + \text{A} \rightarrow \text{Product}$ (slow) $\text{C} + \text{B} \rightarrow \text{product}$
 (C) $\text{A} + \text{A} \rightleftharpoons \text{C}$ (D) $\text{A} + \text{B} \rightleftharpoons \text{C}$
 $\text{B} + \text{B} \rightleftharpoons \text{D}$ $\text{B} + \text{C} \rightarrow \text{D}$ (slow)
 $\text{C} + \text{D} \rightarrow \text{Product}$ (slow) $\text{D} + \text{A} \rightarrow \text{product}$

143. Which straight line gives the activation energy for a reaction?

- (A) rate constant vs T (B) $\ln(\text{rate constant})$ vs T
 (C) rate constant vs T^{-1} (D) $\ln(\text{rate constant})$ vs T^{-1}

144. $\text{CO(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons \text{COCl(g)} + \text{Cl(g)}$ $K_{\text{eq}} = 1.5 \times 10^{-39}$

If the rate constant, k , for the forward reaction above is $1.4 \times 10^{-28} \text{ L}\cdot\text{mol}^{-1}\cdot\text{sec}^{-1}$ what is k (in $\text{L}\cdot\text{mol}^{-1}\cdot\text{sec}^{-1}$) for the backward reaction?

- (A) 2.1×10^{-67} (B) 1.0×10^{-11}
 (C) 9.3×10^{10} (D) 7.1×10^{27}

145. This is the rate law for a reaction that consumes X.

$$\text{rate} = k[\text{X}]^2$$

Which plot gives a straight line?

- (A) $[\text{X}]$ vs. time (B) $\ln[\text{X}]$ vs. time
 (C) $1/[\text{X}]$ vs. time (D) $1/\ln[\text{X}]^2$ vs. time
146. For a first order reaction, the concentration decreases to 30% of its initial value in 5.0 min. What is the rate constant?
 (A) 0.46 min^{-1} (B) 0.24 min^{-1}
 (C) 0.14 min^{-1} (D) 0.060 min^{-1}
147. The rate of a reaction at 75°C is 30.0 times that at 25°C . What is its activation energy?
 (A) $58.6 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $25.5 \text{ kJ}\cdot\text{mol}^{-1}$
 (C) $7.05 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $1.51 \text{ kJ}\cdot\text{mol}^{-1}$
148. $6\text{I}^-(\text{aq}) + \text{BrO}_3^-(\text{aq}) + 6\text{H}^+(\text{aq}) \rightarrow 3\text{I}_2(\text{aq}) + \text{Br}^-(\text{aq}) + 3\text{H}_2\text{O(l)}$
 These data were obtained when this reaction was studied.

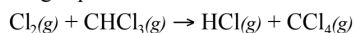
$[\text{I}^-], \text{M}$	$[\text{BrO}_3^-], \text{M}$	$[\text{H}^+], \text{M}$	Reaction rate, $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
0.0010	0.0020	0.010	8.0×10^{-5}
0.0020	0.0020	0.010	1.6×10^{-4}
0.0020	0.0040	0.010	1.6×10^{-4}
0.0010	0.0040	0.020	1.6×10^{-4}

What are the units of the rate constant for this reaction?

- (A) s^{-1} (B) $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
 (C) $\text{L}\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$ (D) $\text{L}^2\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$

Kinetics

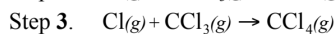
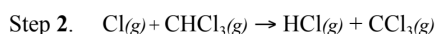
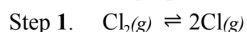
149. Consider this gas phase reaction.



The reaction is found experimentally to follow this rate law.

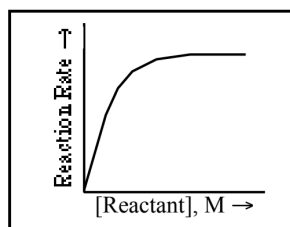
$$\text{rate} = k [\text{CHCl}_3] [\text{Cl}_2]^{1/2}$$

Based on this information, what conclusions can be drawn about this proposed mechanism?



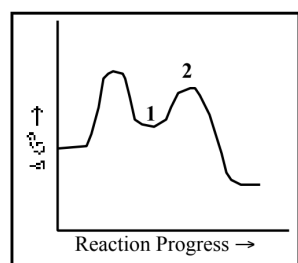
- (A) Step 1 is the rate-determining step.
 (B) Step 2 is the rate-determining step.
 (C) Step 3 is the rate-determining step.
 (D) The rate-determining step cannot be identified.

150. The rates of many catalyzed reactions follow the profile shown in the graph. Why does the reaction rate level off?



- (A) The reactant is used up.
 (B) The reverse reaction becomes dominant.
 (C) The catalyst decomposes as the reaction proceeds.
 (D) The active sites on the catalyst are occupied.

151. What names apply to chemical species corresponding to locations 1 and 2 on this reaction coordinate diagram?

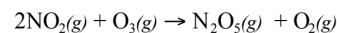


- | Location 1 | Location 2 |
|---------------------------|-----------------------|
| (A) activated complex | activated complex |
| (B) reaction intermediate | activated complex |
| (C) activated complex | reaction intermediate |
| (D) reaction intermediate | reaction intermediate |

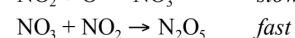
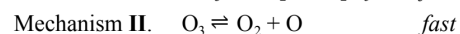
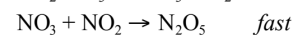
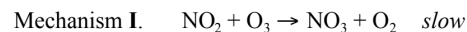
152. Gadolinium-153, which is used to detect osteoporosis, has a half-life of 242 days. Which value is closest to the percentage of the Gd-153 left in a patient's system after 2 years (730 days)?

- (A) 33.0 % (B) 25.0 %
 (C) 12.5 % (D) 6.25 %

153. Consider this reaction.



The reaction of nitrogen dioxide and ozone represented is first order in $\text{NO}_2(\text{g})$ and in $\text{O}_3(\text{g})$. Which of these possible reaction mechanisms is consistent with the rate law?

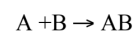


- (A) I only (B) II only
 (C) both I and II (D) neither I nor II

154. When the temperature of a reaction is raised from 300 K to 310 K, the reaction rate doubles. Determine the activation energy, E_a , associated with the reaction.

- (A) 6.45 $\text{kJ} \cdot \text{mol}^{-1}$ (B) 23.3 $\text{kJ} \cdot \text{mol}^{-1}$
 (C) 53.6 $\text{kJ} \cdot \text{mol}^{-1}$ (D) 178 $\text{kJ} \cdot \text{mol}^{-1}$

155. Use the experimental data in the table to determine the rate law for this reaction.



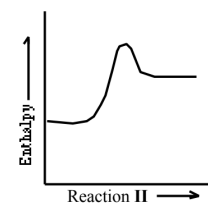
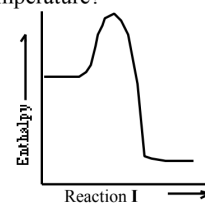
These data were obtained when the reaction was studied.

[A], M	[B], M	$\Delta[\text{AB}]/\Delta t \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$
0.100	0.100	2.0×10^{-4}
0.200	0.100	2.0×10^{-4}
0.300	0.300	1.8×10^{-3}

What is the rate equation for the reaction?

- (A) $\text{rate} = k [\text{A}] [\text{B}]$ (B) $\text{rate} = k [\text{A}]^2$
 (C) $\text{rate} = k [\text{B}]$ (D) $\text{rate} = k [\text{B}]^2$

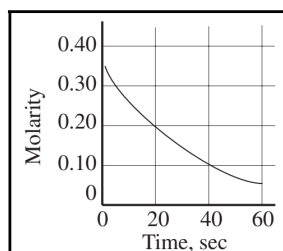
156. Which of the reactions represented in these diagrams will show the greatest increase in rate for a given increase in temperature?



- (A) Reaction I forward (B) Reaction I reversed
 (C) Reaction II forward (D) Reaction II reversed

Kinetics

157. A reaction follows this concentration-time diagram. The instantaneous rate for this reaction at 20 seconds will be closest to which value?

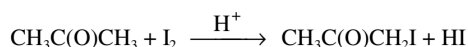


- (A) $4 \times 10^{-3} \text{ M}\cdot\text{sec}^{-1}$ (B) $8 \times 10^{-3} \text{ M}\cdot\text{sec}^{-1}$
 (C) $2 \times 10^{-2} \text{ M}\cdot\text{sec}^{-1}$ (D) $1 \times 10^{-1} \text{ M}\cdot\text{sec}^{-1}$
158. If the half-life of a reaction increases as the initial concentration of substance increases, the order of the reaction is
- (A) zero. (B) first.
 (C) second. (D) third.

159. The radioisotope N-13, which has a half-life of 10 minutes, is used to image organs in the body. If an injected sample has an activity of 40 microcuries (40 μCi), what is its activity after 25 minutes in the body?

- (A) 0.75 μCi (B) 3.5 μCi
 (C) 7.1 μCi (D) 12 μCi

160. Propanone reacts with iodine in acid solution as shown in this equation.



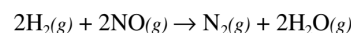
These data were obtained when the reaction was studied.

$[\text{CH}_3\text{C}(\text{O})\text{CH}_3], \text{M}$	$[\text{I}_2], \text{M}$	$[\text{H}^+], \text{M}$	Relative Rate
0.010	0.010	0.010	1
0.020	0.010	0.010	2
0.020	0.020	0.010	2
0.020	0.010	0.020	4

What is the rate equation for the reaction?

- (A) $\text{rate} = k[\text{CH}_3\text{C}(\text{O})\text{CH}_3][\text{I}_2]$
 (B) $\text{rate} = k[\text{CH}_3\text{C}(\text{O})\text{CH}_3]^2$
 (C) $\text{rate} = k[\text{CH}_3\text{C}(\text{O})\text{CH}_3][\text{I}_2][\text{H}^+]$
 (D) $\text{rate} = k[\text{CH}_3\text{C}(\text{O})\text{CH}_3][\text{H}^+]$
161. A particular reaction rate increases by a factor of five when the temperature is increased from 5 °C to 27 °C. What is the activation energy of the reaction?
- (A) 6.10 $\text{kJ}\cdot\text{mol}^{-1}$ (B) 18.9 $\text{kJ}\cdot\text{mol}^{-1}$
 (C) 50.7 $\text{kJ}\cdot\text{mol}^{-1}$ (D) 157 $\text{kJ}\cdot\text{mol}^{-1}$

162. Consider this reaction.



The rate law for this reaction is $\text{rate} = k[\text{H}_2][\text{NO}]^2$. Under what conditions could these steps represent the mechanism?

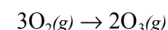
- Step 1. $2\text{NO} \rightleftharpoons \text{N}_2\text{O}_2$
 Step 2. $\text{N}_2\text{O}_2 + \text{H}_2 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$
 Step 3. $\text{N}_2\text{O} + \text{H}_2 \rightarrow \text{N}_2 + \text{H}_2\text{O}$

- (A) These steps cannot be the mechanism under any circumstances.
 (B) These steps could be the mechanism if step 1 is the slow step.
 (C) These steps could be the mechanism if step 2 is the slow step.
 (D) These steps could be the mechanism if step 3 is the slow step.

163. A reaction has a forward rate constant of $2.3 \times 10^6 \text{ s}^{-1}$ and an equilibrium constant of 4.0×10^8 . What is the rate constant for the reverse reaction?

- (A) $1.1 \times 10^{-15} \text{ s}^{-1}$ (B) $5.8 \times 10^{-3} \text{ s}^{-1}$
 (C) $1.7 \times 10^2 \text{ s}^{-1}$ (D) $9.2 \times 10^{14} \text{ s}^{-1}$

164. The rate of formation of $\text{O}_3(\text{g})$ is $2.0 \times 10^{-7} \text{ mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$ for the reaction



What is the rate of disappearance of $\text{O}_2(\text{g})$ in $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$?

- (A) 1.3×10^{-7} (B) 2.0×10^{-7}
 (C) 3.0×10^{-7} (D) 4.5×10^{-7}

165. The reaction between $\text{NO}(\text{g})$ and $\text{O}_2(\text{g})$ to give $\text{NO}_2(\text{g})$ is second order in $\text{NO}(\text{g})$ and first order in $\text{O}_2(\text{g})$. By what factor will the reaction rate change if the concentrations of both reactants are doubled?

- (A) 2 (B) 4 (C) 6 (D) 8

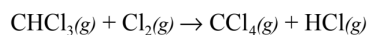
166. The decomposition of ethane into two methyl radicals has a first order rate constant of $5.5 \times 10^{-4} \text{ sec}^{-1}$ at 700 °C. What is the half-life for this decomposition in minutes?

- (A) 9.1 (B) 15 (C) 21 (D) 30

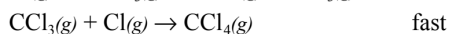
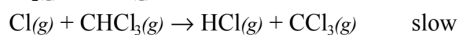
167. The dependence of the rate constant of a reaction on temperature is given by the equation $k = e^{-E_a/kT}$. Under what conditions is k the smallest?

- (A) high T and large E_a (B) high T and small E_a
 (C) low T and large E_a (D) low T and small E_a

168. The reaction



is believed to proceed by this mechanism:



What rate equation is consistent with this mechanism?

(A) $\text{Rate} = k[\text{Cl}_2]$

(B) $\text{Rate} = k[\text{Cl}][\text{CHCl}_3]$

(C) $\text{Rate} = k[\text{Cl}_2][\text{CHCl}_3]$

(D) $\text{Rate} = k[\text{Cl}_2]^{1/2}[\text{CHCl}_3]$

169. The activation energy of a certain reaction is $87 \text{ kJ} \cdot \text{mol}^{-1}$.
What is the ratio of the rate constants for this reaction
when the temperature is decreased from 37°C to 15°C ?

(A) 5/1 (B) 8.3/1 (C) 13/1 (D) 24/1